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National Institute of Standard and Technology
Gaithersburg, MD, Building 215
Cleanroom Re-Balance Phasing

(100% Submission)

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Overview

On June 29, 2021, Burdette, Koehler, Murphy, and Associates (BKM) issued construction documents to the National Institute of Standards and Technology (NIST) for the re-balancing of the Cleanrooms in Building 215. Although the design documents were completed and issued, funding for construction has not previously been released. The original intent assumed that the Cleanrooms could be shut down for an extended period and therefore did not include a construction phasing plan. However, NIST has since clarified that extended shutdowns are not permissible, and that construction must be phased to minimize disruption to ongoing Cleanroom operations. The purpose of this project is to develop a construction phasing plan to supplement the Cleanroom Re-Balance design documents.



Typical Galley



Typical Cleanroom

Cleanroom Re-Balance Design Scope

The Cleanroom Re-Balance design documents for Building 215 include the following scope items, refer to the design documents dated June 29th, 2021, for more detailed information.

1. *HEPA filters:* New HEPA terminal filters & housing will be installed within the Class 100 and 1000 spaces to replace the existing original filters & housings. The existing original HEPAs are integral to the module and the entirety of the module will be replaced and reinstalled into the ceiling assembly. The contractor shall confirm during demolition if gasketing or gel assembly is used for the existing original HEPA filters. The gasketing or gel assembly shall be replaced or repaired to ensure proper sealing at ceiling assembly to maintain proper certification of each HEPA filter. The AHU HEPA filters in the recirculating and make-up air units are replaced under a normal preventative maintenance schedule and will be completed under regularly scheduled maintenance. The replacement of the filter will allow the associated recirculating AHU fans to operate at a reduced fan speed and reduce the overall fan energy usage. The quantity and locations of the existing HEPA devices are indicated on Drawing 215-M102.
2. *VAV terminals:* If it is found that the existing Nailor VAV terminals controllers have begun to fail, these controllers will be replaced or if funding is available the terminal units are replaced in kind. Equipment plans indicate air terminal units to be tested and schedules to indicate final balancing airflow values.
3. *Replace AFMs:* The existing Ebtron devices and readouts will be replaced with new airflow measuring stations and local readout stations. As part of this replacement the existing thirteen branch dampers on the solvent, acid and hydride systems will be replaced with new pressure independent air valve, such as Accu-valve or Phoenix exhaust air terminals. The new exhaust valves will allow each branch duct to operate off of pressure in the ductwork and modulate the damper to maintain flow if a tool is removed or shuts off due to intermittent use. The airflow measuring stations are indicated on the airflow schematic and are located on a part plan view typical of Drawing 215-M201.
4. *Replace magnehelic gauges:* The existing magnehelic gauges installed on EF-102 through EF-105 and the Cleanrooms will be replaced. The exhaust gauges will be installed with an appropriate operating range for each fan. The replacement magnehelic gauges installed within the Cleanroom will utilize a $-0.25''$ wc to $+0.25''$ w.c. tolerance to indicate if the Class 100 space drops to a neutral or negative pressurization. All gauges will be tied to the BAS for trending, monitoring and alert purposes.
5. *Repair ductwork:* Each exhaust tool shall be installed with a volume damper, including replacing existing failing dampers, to create a uniform system that can be rebalanced when new equipment is installed within a lab. Typical repair of failing joints or connections are identified to ensure the safety of Cleanroom operators and staff. All existing exhaust tools have not verified since the initial release of the 2021 construction documents; the contractor shall verify existing conditions and proposed equipment balancing prior to the start of work. Prior to the start of each phase the contractor shall submit a phasing plan, including airflow balancing and pressurization plans (during and post-construction) for Government & Engineers review and approval.
6. *Return Air Pathways:* Due to a large quantity of air that is collected along Corridor CO-D-1 a wind tunnel effect is created in this area. A new above ceiling return air transfer will be installed to allow air to migrate back to AHU-113 & AHU-115 freely without entering Room D107. This will relieve air transfer across the door, allowing it to open and close freely. Room D107 is designated as a chemical change-out room. This space is recommended to be exhausted rather than allowing air to circulate back to other adjacent spaces. The existing supply air within Room D107 supplies 120 air changes per hour, so ductwork modifications will be required to exhaust the air within Room D107 to the outdoors rather than allowing it to migrate to the return of AHU-113.

7. *Testing of AHUs / EFs*: It is expected that several spot check locations will be completed to determine if an existing sensor or sensors are accurate or there is a reporting issue with the BAS. Equipment plans indicate equipment to be tested, and schedules indicate final balancing airflow values.
8. *Calibration & Verification of AFMs*: As previously stated under the “replace AFM” section a testing balancing contractor will first verify that the existing Ebtron airflow stations are not accurately reading the airflow and will be replaced in kind as required. It is recommended that initial testing and investigation be completed prior to replacement to ensure that a newly installed device maintains its expected tolerance and longevity.
9. *Rebalance of Cleanroom*: The Building 215 air systems will be rebalanced. The equipment to be included in this rebalance is as listed: AHU-101A through 101T, AHU-102A through 102D, AHU-103, AHU-104, AHU-108A, AHU-108B, AHU-109, AHU-110A, AHU-110B, AHU-111A through 111D, AHU-112, AHU-113, AHU-114, AHU-115, EF-101A, EF-101B, EF-102A, EF-102B, EF-105A, EF-105B & EF-114. Airflows for each piece of lab tool and HVAC equipment will be rebalanced based on flows indicated on the construction documents or pre-demolition airflow readings.
10. *Correct airflow in lobby and other exhaust systems*: The Building 215 air systems will be rebalanced. The equipment to be included in this rebalance is as listed: AHU-105, AHU-106, AHU-107, AHU-123, EF-103, EF-104, EF-106, EF-108, EF-109, EF-112, EF-122, EF-123, RF-105 & RF-107.

As a part of the Cleanroom Re-Balance design documents, there are additional modifications that are considered to improve the existing Cleanroom and adjacent spaces operation.

1. Ensure a consistent preventative maintenance schedule is in place to replace the HEPA and carbon filters located in the recirculating and make-up air units.
2. Install differential pressure sensors across a select number of Cleanroom terminal HEPA filters to alert the BAS due to an increase in pressure drop across the filter, indicating the filters are loaded and requiring changing. This will allow for filter loading and fan energy to be monitored and trended.
3. Install differential pressure sensors at the Class 100 doors to alert the BAS that a space has dropped below a specified pressure differential to prevent any potential transfer of dirt or dust that may damage the on-going research.
4. The Cleanroom doors between the Class 1000 and viewing corridor are not gasketed. If these doors and other openings were sealed, it would ensure additional pressurization within the Cleanroom, which would allow for an overall reduction in make-up air required. As 100% make-up air requires heating and cooling, a reduction in the usage of make-up air would provide significant energy savings for the Cleanroom operation.
5. Commissioning services shall be provided after all modifications and re-balancing activities have been completed to verify that the air systems are indeed operating as the construction documents indicate.

Cleanroom Construction Phasing Plan

The Cleanroom facility comprises 14 individual Cleanroom bays, each connected via equipment galleys, corridors, gowning areas, and other support spaces. NIST has requested that each Cleanroom Bay be isolated and shut down individually during each phase of construction, allowing the remaining bays to remain operational. This phased approach will continue until construction is completed in all Cleanroom areas. Each construction phase will include but is not limited to the following items, refer to the original design documents for additional information:

1. Test associated recirculating AHUs
2. Perform equipment modifications to associated AHUs
3. Provide protective covering of all existing lab equipment and provisions to provide access to HEPA filters and HEPA housing connections
4. Isolate Cleanroom Bay with temporary partitions
5. Shutdown the associated AHUs serving the Cleanroom Bay and close associated make up air dampers
6. Shutdown exhaust tools within the Cleanroom Bay and modify exhaust ductwork as required
7. Replace associated HEPA filters & housing in Cleanroom Bay
8. Replace associated magnehelic gauges
9. Replace associated AFMs
10. Rebalance associated supply and exhaust air systems
11. Perform HEPA certification
12. Perform Cleanroom recertification

Due to the mission critical work and research that is conducted in the Building 215 NanoFAB cleanroom, the ability to perform work in a safe and precise manner is a key requirement for the success of this project.

Refer to the following additional guidelines and parameters for listed assumptions:

- A. Construction may take place 24 hours per day and 7 days a week for the duration of prescribed outages.
- B. Outages/Phases are anticipated to be scheduled in coordination with holidays to minimize operational impacts to work inside the Cleanroom
- C. Multiple construction crews will be anticipated during each shift, however contractors shall consider where additional personnel can be provided to accelerate activities.
- D. Where activities are completed ahead of anticipated durations, it is expected that the next task will commence immediately to shorten the overall outage associated with the area of work.
- E. Task durations should include staging, gowning, personnel breaks, or contractor badging/check-in. All staging and associated work shall be coordinated and scheduled to complete scope of work.
- F. Where modifications to the existing exhaust ductwork or tools are required as part of the scope of work, it is anticipated that this work is scheduled and completed prior to or incorporated into each phases scheduled outage.
- G. All existing exhaust tools have not verified since the initial release of the 2021 construction documents; the contractor shall verify existing conditions and proposed equipment balancing prior to the start of work. Prior to the start of each phase the contractor shall submit a phasing plan, including airflow balancing and pressurization plans (during and post-construction) for Government & Engineers review and approval.

Below are some baseline tasks to be included within each phase of work.

1. Setup scaffolding, laboratory equipment protection, and temporary partitions.
2. HEPA filter & housing replacement:
 - a. Construction crew(s) will be assigned to remove the HEPA devices
 - b. Construction crew(s) will be assigned to remove the gel gasketing, clean the construction space, and do miscellaneous tasks
 - c. Construction Crew(s) will be assigned to install the new HEPA devices.
3. Balancing:
 - a. Rebalance AHUs
 - b. Rebalance HEPA devices and room pressurization
4. HEPA certification
5. Remove scaffolding, laboratory equipment protection, and temporary partitions
6. Final cleaning of space
7. Area of work cleanroom certification

Please note there will be additional time needed for miscellaneous requirements and work outside of the cleanroom area, refer to the original design documents for additional information.

Phases are anticipated to be completed in the following order or based on written approval or order and scheduling as approved by NIST COR and NanoFAB cleanroom representative.

1. Class 1000 Corridor A
2. Class 1000 Corridor B
3. Central Corridor
4. Cleanrooms Module
 - 4.1. B107 & B106-2
 - 4.2. B101
 - 4.3. B102
 - 4.4. B103
 - 4.5. B104
 - 4.6. B105
 - 4.7. B106
 - 4.8. A106
 - 4.9. A105
 - 4.10. A104
 - 4.11. A103
 - 4.12. A102
 - 4.13. A101
5. D106, D107 & D108
6. D101 & D105
7. Non-Critical Spaces

Appendix One presents a visual representation of the anticipated construction sequence as requested by NIST. Based on the request from the Cleanroom personnel, it is anticipated that there will be approximately 19 construction phases.

Below is summary of each construction phase:

1. Class 1000 Corridor A (Phase 1; Refer to Appendix One for location)
 - a. General Construction Sequencing:
 - i. Corridor A could be split into two sub-phases to minimize disruptions to the lithography modules (A101, A102, A103, A104, A105 & A106)
 - ii. This space can be completed anytime in relation to all other spaces
 - b. General Construction Scope:
 - i. Approximately 84 HEPA filters
 - ii. No anticipated exhaust tools
 - iii. Refer to construction documents for additional information
 - c. General Construction Tasks:
 - i. Setup scaffolding and laboratory equipment protection
 - ii. Remove and replace HEPA devices
 - iii. Balance HEPA devices
 - iv. Certify HEPA devices
 - v. Remove scaffolding and laboratory equipment protection
 - vi. Final cleaning of space
 - vii. Cleanroom certification



Corridor A

2. Class 1000 Corridor B (Phase 2; Refer to Appendix One for location)

a. General Construction Sequencing:

- i. Cleanroom Bays B101 and B102 contain acid and share an emergency eye wash/shower station located in class 1000 Corridor B
- ii. Access to the existing emergency eyewash/shower or provisions for temporary emergency eyewash/shower shall be provided and comply with ANSI Z358.1 requirements
- iii. This space can be completed anytime in relation to all other spaces

b. General Construction Scope:

- i. Approximately 84 HEPA filters
- ii. No anticipated exhaust tools
- iii. Refer to construction documents for additional information

c. General Construction Tasks:

- i. Setup scaffolding and laboratory equipment protection
- ii. Remove and replace HEPA devices
- iii. Balance HEPA devices
- iv. Certify HEPA devices
- v. Remove scaffolding and laboratory equipment protection
- vi. Final cleaning of space
- vii. Cleanroom certification



Corridor B

3. Central Corridor (Phase 3; Refer to Appendix One for location)

- a. General Construction Sequencing:
 - i. This space can be completed anytime in relation to all other spaces
 - ii. Temporary partitions will need to be provided to isolate the central corridor from the “B” side Cleanrooms
- b. General Construction Scope:
 - i. Approximately 88 HEPA filters
 - ii. No anticipated exhaust tools
 - iii. Refer to construction documents for additional information
- c. General Construction Tasks:
 - i. Setup scaffolding and laboratory equipment protection
 - ii. Remove and replace HEPA devices
 - iii. Balance HEPA devices
 - iv. Certify HEPA devices
 - v. Remove scaffolding and laboratory equipment protection
 - vi. Final cleaning of space
 - vii. Cleanroom certification



Central Corridor

4. Cleanrooms A101, A102, A103, A104, A105, A106, B101, B102, B103, B104, B105, B106, B106-2, B107 (Phases 4.1-4.13; Refer to Appendix One for locations)
 - a. General Construction Sequencing:
 - i. Cleanrooms shall be completed one at a time so that each Cleanroom is a single outage
 - ii. Cleanroom A101 will be the last cleanroom in this sequence to be completed to ensure workflow, protocols and standard of care have been maximized as this space includes essential equipment.
 - iii. Cleanroom B107 and B106-2 will be combined into one phase due to HEPA devices that straddle the wall between the spaces and serve both spaces.
 - b. General Construction Scope:
 - i. Approximately 70 HEPA filters per Cleanroom
 - ii. Approximately 8 exhaust tools per Cleanroom
 - iii. Refer to construction documents for additional information
 - c. General Construction Tasks (per cleanroom):
 - i. Setup scaffolding and laboratory equipment protection
 - ii. Remove and replace HEPA devices
 - iii. Balance HEPA devices
 - iv. Certify HEPA devices
 - v. Remove scaffolding and laboratory equipment protection
 - vi. Final cleaning of space
 - vii. Cleanroom certification



Typical Cleanroom Bay



Typical Cleanroom Bay

5. Spaces D106, D107, and D108 (Phase 5; Refer to Appendix One for locations)

- a. General Construction Sequencing:
 - i. These spaces are require additional ductwork modification and are smaller spaces that will limit the number of personnel capable of working within this area simultaneously.
- b. General Construction Scope:
 - i. Approximately 21 HEPA filters
 - ii. Approximately 3 exhaust tools
 - iii. Ductwork modifications above ceiling
 - iv. Refer to construction documents for additional information
- c. General Construction Tasks:
 - i. Setup scaffolding and laboratory equipment protection
 - ii. Remove and replace HEPA devices
 - iii. Balance HEPA devices
 - iv. Certify HEPA devices
 - v. Remove scaffolding and laboratory equipment protection
 - vi. Final cleaning of space
 - vii. Cleanroom certification

6. Spaces D101 and D105 (Phase 6; Refer to Appendix One for locations)

- a. General Construction Sequencing:
 - i. These spaces are essential to the operation of the Cleanroom as these spaces are used for gowning and entering the Cleanroom
 - ii. These spaces cannot have an outage during normal operating hours
 - iii. These spaces can be completed anytime in relation to all other spaces
- b. General Construction Scope:
 - i. Approximately 33 HEPA filters
 - ii. No anticipated exhaust tools
 - iii. Refer to construction documents for additional information
- c. General Construction Tasks:
 - i. Setup scaffolding and laboratory equipment protection
 - ii. Remove and replace HEPA devices
 - iii. Balance HEPA devices
 - iv. Certify HEPA devices
 - v. Remove scaffolding and laboratory equipment protection
 - vi. Final cleaning of space
 - vii. Cleanroom certification

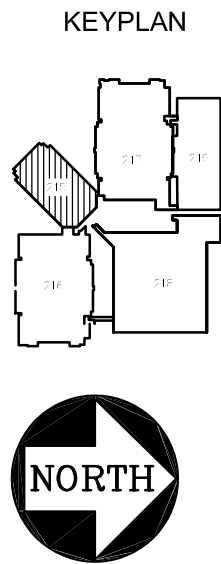
7. Non-critical spaces

- a. All other work as listed in the scope of work or construction documents is anticipated to be conducted during normal business hours, unless associated work or outage impacts operations of the Cleanroom. Where work does impact Cleanroom activities or operations, contractor shall assume work to be completed on weekends or holidays unless otherwise approved by NIST COR.

Conclusion

In conclusion, the Building 215 Cleanroom Re-Balance project provides a comprehensive approach to improving airflow performance, energy efficiency, system reliability, and operational control within the NanoFAB facility. While the original 2021 design assumed a single extended shutdown, the updated construction phasing plan responds to NIST's requirement to minimize cleanroom outages by isolating and upgrading one cleanroom bay at a time. Through coordinated HEPA filter replacements, airflow monitoring upgrades, ductwork modifications, system rebalancing, and full recertification, each phase is structured to minimize disruption while maintaining safety, cleanliness standards, and research continuity. The contractors will be responsible for careful scheduling, testing, and commissioning, for each phase of the project. It will be the contractor's responsibility to best meet the prescribed schedule requirements while protecting laboratory equipment and operations within the NanoFAB Cleanroom. It is expected that all work will be scheduled in close coordination with the government and laboratory end users for any work occurring inside the cleanroom or having an impact on the operations of the cleanroom.

Appendix One – Construction Phases



1. PERFORM AS-FOUND TESTING OF EQUIPMENT OF AIRFLOW, FAN AND SYSTEM COMPONENTS. REBALANCE TO SCHEDULED AIRFLOW.
2. INSTALL NEW 800x800 DUCTWORK, REFER TO DETAIL K3/215-512.
3. DISCONNECT EXISTING 75 EXHAUST DOWN TO ACID STORAGE CABINET, RECONNECT TO NEW 400 EXHAUST AIR DUCT. BALANCE TO 30 L/S.
4. PERFORM AS-FOUND AIRFLOW AND FUNCTION TEST ON EXISTING TERMINAL UNIT. REPORT FINDINGS TO CORP AND ENGINEER. UNDER BASE BID, EXISTING TERMINAL UNIT CONTROLLER TO BE REPLACED WITH COMPATIBLE CONTROLLER FOR EXISTING NAILOR TERMINAL UNIT. EXISTING TERMINAL UNIT TO REMAIN AND CLEANED IN PLACE, INCLUDING AIRFLOW MEASUREMENT ELEMENTS. TEMPORARILY DISCONNECT DUCTWORK FROM TERMINAL UNIT AS REQUIRED TO COMPLETE CLEANING, AND RECONNECT AND SEAL DUCTWORK AFTER COMPLETION OF WORK. CONTROL TUBING SHALL BE REPLACED AND EXTENDED AS REQUIRED TO BE CONNECTED TO EXISTING BUILDING AUTOMATION SYSTEM. ALL NEW CONTROLS SHALL BE VERIFIED FOR ACCURACY AND RESPONSE. UNDER ALTERNATE 1, REPLACE EXISTING TERMINAL WITH A NEW SUPPLY AIR TERMINAL UNIT TO MATCH EXISTING. RECONNECT TO EXISTING DUCTWORK AND HVAC PIPING. CONTROL TUBING SHALL BE REPLACED AND NEW CONTROLS SHALL BE EXTENDED TO EXISTING BUILDING AUTOMATION SYSTEM. TYPICAL OF 25.
5. EXISTING TO REMAIN 300x150 BACKDRAFT DAMPER IN WALL ABOVE CEILING (100 L/S).
6. INSTALL NEW 800x800 RETURN AIR TRANSFER GRILLE IN WALL ABOVE CEILING. REFER TO DETAIL.
7. EXISTING TO REMAIN 350x200 TRANSFER AIR DUCT.
8. EXISTING TO REMAIN 300x150 TRANSFER AIR DUCT.
9. EXISTING TO REMAIN THERMOSTAT. RECONNECT TO NEW TERMINAL UNIT CONTROLLER.
10. EXISTING FAN COIL UNIT TO REMAIN. NOT IN SCOPE OF WORK.
11. 950x400 EXHAUST AIR DUCT UP TO EF-103.
12. 1550x400 EXHAUST AIR DUCT UP TO EF-104.
13. NEW 300 WELDED STAINLESS STEEL EXHAUST AIR DUCT. BALANCE TO 560 L/S.
14. NEW 400 WELDED STAINLESS STEEL EXHAUST AIR DUCT. REFER TO 215-490 FOR CONTINUATION.
15. 75 EXHAUST DOWN TO ACID STORAGE CABINET. BALANCE TO 30 L/S.

SHEET ID
215-467